

Claire Kim

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Education

University of California, Los Angeles (UCLA) | Bachelor of Science in Mechanical Engineering **Sep 2025 – Jun 2029**

Projects

Mechanical Engineering Lead

Apr 2026 - Present

Petalux | Creative Labs

Los Angeles, CA

- Engineered the complete mechanical structure of an automated kinetic lamp in **Autodesk Fusion 360**, driving the project from initial design requirements to fully constrained 3D CAD assemblies and physical prototypes
- Designed a custom powertrain featuring a central screw linkage and specialized petal gears to actuate a 10-petal opening sequence while minimizing axial wobble and gear shifting through tight-tolerance design
- Developed space-efficient internal mounting structures and routing channels within the chassis to seamlessly house micro servos, wiring, and electrical components without obstructing moving mechanical parts
- Partnered with cross-functional design and electronics tracks to finalize the chassis enclosure, integrating user-interface hardware via internal snap-fits, speaker housings, and custom filleted screen cutouts

Mechanical Engineering Lead

Oct 2025 - Nov 2025

Aeris | Theta Tau

Los Angeles, CA

- Directed “Aeris”, a smart-cooling system using facial recognition with tri-axis actuation for automated airflow
- Engineered custom enclosures and complex mechanical linkages in SolidWorks and Fusion 360, prioritizing Design for Assembly (DFA) to streamline the integration of Arduino hardware via high-precision friction fitting
- Managed full product lifecycle from FDM 3D printing to system validation, performing precision soldering, electromechanical assembly, and iterative troubleshooting to resolve hardware errors

Chassis Mechanical Engineer

Sept 2025 - Present

BB-8 | X1 Robotics - American Society of Mechanical Engineers

Los Angeles, CA

- Engineered and assembled BB-8 mechanisms, optimizing balance, motion, and overall mechanical functionality
- Utilized SolidWorks to design drivetrain and internal arm components, positioning motor housings to fit the shell and minimizing structures to reduce weight while ensuring precise robot movement and machinability
- Developed core BB-8 support ring in CAD, factoring stress tolerance and in-house manufacturing feasibility

Experience

RCMakes

Mar 2022 - Jul 2025

Lead Intern & Jobshop Lead

Camarillo, CA

- **Digital Fabrication & Mentorship:** Managed 80+ interns in digital fabrication workflows, providing technical instruction on CNC machining, laser cutting, and 3D printing to generate \$5k in total shop revenue
- **Technical Operations & Outreach:** Developed an Arduino robotics curriculum for STEM workshops and directed advanced theater technology, including lighting, sound, and projection, for 500-seat community events

UCLA COSMOS

Jul 2024 - Aug 2024

Student in Cluster 5: Exploring Mechanical and Aerospace Engineering Through Mechatronics

Los Angeles, CA

- **Robotics & Control Systems:** Constructed and programmed an autonomous, self-balancing two-wheeled robot in C++ and Arduino; tuned PID controls and utilized ultrasonic sensing for high-speed maze navigation
- **Rocketry:** Applied OpenRocket and fluid dynamics to design and optimize a model rocket for peak flight stability

Leadership

WEMech | Club & Nonprofit

Aug 2022 - Aug 2025

Nonprofit Co-Founder / President / CFO / Secretary / Agent of Service

Camarillo, CA

- Partnered with 10+ schools to teach engineering workshops to students through immersive STEM activities
- Mentored 100+ girls in hands-on engineering projects and leadership growth, fostering confidence in STEM
- Managed WEMech’s nonprofit registration and raised \$3,000+ to support events and organizational operations

Technical Skills

- SolidWorks, Fusion360, Arduino, AutoCAD, Soldering, Machining, Welding, OpenRocket
- Programming Languages: Python, C++, JavaScript, CSS/HTML
- Relevant Coursework: Engineering Design/CAD, MATLAB Programming